

Problem Set 3

Fundamentals of Numerical Programming I

LEONARD STORCKS

January 6, 2026

Problem 1. Numerical derivatives by finite differencing

Todo

- (a) Determine the coefficients a and b in the symmetric Ansatz

$$f'(x) = a(f(x + \Delta x) - f(x - \Delta x)) + b(f(x + 2\Delta x) - 2f(x - \Delta x)) + \mathcal{O}(\Delta x^4) \quad (1)$$

to obtain a 4-th order finite difference approximation of $f'(x)$ by Taylor expansion.

- (b) Implement Python functions for the

- first order forward differencing formula
- second order central differencing formula
- fourth order differencing formula

- (c) For $f(x) = \sin(x)$ and $x_i = \dots$ plot the error over Δx (see paper)

- (d) Explain the behavior you observe.

Problem 2. Newton interpolation and Runge's phenomenon

Todo

(a) todo

Problem 3. The power of finite differencing - solving the Kuramoto–Sivashinsky (KS) equation

Todo

(a) todo